

# More HW problems (for those who can't get enough...)

## EASY PROBLEMS

## ANSWER KEY

For the first three problems calculate the frequency of the three genotypes and then calculate the frequency of the two phenotypes.

1. A gaggle of geese with  $p = 0.25$ .

$$p = 0.25, q = 0.75 \quad (0.25 + 0.75 = 1.0)$$

$$p^2 = 0.0625$$

$$AA = 0.0625$$

$$2pq = 0.375$$

$$Aa = 0.375$$

$$q^2 = 0.5625$$

$$aa = 0.5625$$

$$\text{CHECK: } (0.0625 + 0.375 + 0.5625 = 1.0)$$

$$\text{"A\_"} \text{ phenotype} = 0.0625 + 0.375 = 0.4375$$

$$\text{"a"} \text{ phenotype} = 0.5625$$

$$\text{CHECK: } (0.4375 + 0.5625 = 1.0)$$

2. A school of fish with  $p = 0.65$ .

$$p = 0.65, q = 0.35 \quad (0.65 + 0.35 = 1.0)$$

$$p^2 = 0.4225$$

$$AA = 0.4225$$

$$2pq = 0.455$$

$$Aa = 0.455$$

$$q^2 = 0.1225$$

$$aa = 0.1225$$

$$\text{CHECK: } (0.4225 + 0.455 + 0.1225 = 1.0)$$

$$\text{"A\_"} \text{ phenotype} = 0.4225 + 0.455 = 0.8775$$

$$\text{"a"} \text{ phenotype} = 0.1225$$

$$\text{CHECK: } (0.8775 + 0.1225 = 1.0)$$

3. A herd of elephants with  $q = 0.65$ .

$$p = 0.35, q = 0.65 \quad (0.35 + 0.65 = 1.0)$$

$$p^2 = 0.1225$$

$$AA = 0.1225$$

$$2pq = 0.455$$

$$Aa = 0.455$$

$$q^2 = 0.4225$$

$$aa = 0.4225$$

$$\text{CHECK: } (0.1225 + 0.455 + 0.4225 = 1.0)$$

$$\text{"A\_"} \text{ phenotype} = 0.1225 + 0.455 = \mathbf{0.5775}$$

$$\text{"a"} \text{ phenotype} = \mathbf{0.4225}$$

$$\text{CHECK: } (0.5775 + 0.4225 = 1.0)$$

For the next three problems calculate the frequency of the three genotypes. Also calculate the hypothesized actual number of the genotypes and the phenotypes.

4. A herd of 300 goats with 12 homozygous recessive individuals.

$$p=0.8, q=0.2 \text{ (} 0.8 + 0.2 = 1.0 \text{)}$$

$$p^2 = 0.64$$

$$\mathbf{AA = 0.64}$$

$$2pq = 0.32$$

$$\mathbf{Aa = 0.32}$$

$$q^2 = 0.04$$

$$\mathbf{aa = 0.04}$$

$$\text{CHECK: } (0.64 + 0.32 + 0.04 = 1.0)$$

$$AA = 300 \times 0.64 = 192$$

$$Aa = 300 \times 0.32 = 96$$

$$aa = 300 \times 0.04 = 12$$

$$\text{"A\_"} \text{ phenotype} = 192 + 96 = \mathbf{288}$$

$$\text{"a"} \text{ phenotype} = \mathbf{12}$$

$$\text{CHECK: } (288 + 12 = 300)$$

5. A galaxy of 86 sea stars (starfish) includes 24 individuals which have a recessive skin disorder causing them to be fluorescent pink. Their normal color is gray.

$$p=0.472, q=0.528 \text{ (} 0.472 + 0.528 = 1.0 \text{)}$$

$$p^2 = 0.223$$

$$\mathbf{AA = 0.223}$$

$$2pq = 0.498$$

$$\mathbf{Aa = 0.498}$$

$$q^2 = 0.279$$

$$\mathbf{aa = 0.279}$$

$$\text{CHECK: } (0.223 + 0.498 + 0.279 = 1.0)$$

$$AA = 86 \times 0.223 = 19.18 = 19$$

$$Aa = 86 \times 0.498 = 42.83 = 43$$

$$aa = 86 \times 0.279 = 23.99 = 24$$

$$\text{"A"} \text{ (gray) phenotype} = 19 + 43 = \mathbf{62}$$

$$\text{"a"} \text{ (fluorescent pink) phenotype} = \mathbf{24}$$

$$\text{CHECK: } (62 + 24 = 86)$$

6. A flock of 125 scarlet canaries (a type of bird) that includes 15 individuals with gold colored feathers (a recessive mutation).

$$p=0.654, q=0.346 \quad (0.654 + 0.346 = 1.0)$$

$$p^2 = 0.427$$

$$AA = 0.427$$

$$2pq = 0.453$$

$$Aa = 0.453$$

$$q^2 = 0.12$$

$$Aa = 0.12$$

$$\text{CHECK: } (0.427 + 0.453 + 0.12 = 1.0)$$

$$AA = 125 \times 0.427 = 53.4 = 53$$

$$Aa = 125 \times 0.453 = 56.6 = 57$$

$$aa = 125 \times 0.12 = 15$$

$$\text{"A" phenotype} = 53.4 + 56.6 = 110$$

$$\text{"a" phenotype} = 15$$

$$\text{CHECK: } (110 + 15 = 125)$$

For this problem you are making a prediction about the next generation.

7. A flock of 68 scarlet tanagers, 14 of which have a white cap (a recessive mutation). They breed and successfully raise 94 chicks. How many of these chicks do you expect to have a white cap?

$$14/68 = 0.206$$

$$q^2 = 0.206 \text{ or } 20.6\% \text{ have a white cap}$$

$$aa = 94 \text{ chicks} \times 0.206 = 19.4 = \text{19 chicks should have a white cap}$$

## MORE DIFFICULT PROBLEMS

8. You find a flock of birds in the wild. There are 120 with orange bills and 63 with brown bills. When you did breeding experiments, you found that the orange billed birds were true-breeding and the brown billed birds did NOT breed true. Based on that, calculate the values of  $p$ ,  $p^2$ ,  $2pq$ ,  $q$  and  $q^2$ .

For this problem, you need to know what the recessive phenotype is. Because the orange billed birds “true breed” (meaning when they are self-crossed, they produce offspring ONLY like themselves and have orange bills), they HAVE to be homozygous recessive or “aa”.

$$q^2 = 120 \text{ orange bills} / 183 \text{ total birds} = 0.656$$

$$q = 0.810$$

$$p = 0.190$$

$$p^2 = 0.036$$

$$2pq = 0.308$$

$$\text{CHECK: } (0.036 + 0.308 + 0.656) = 1.0$$

9. You find a patch of flowers growing in the wild. There are 3428 with blue petals and 1853 with black petals. When you did breeding experiments in your greenhouse, you found that black petal flowers were true-breeding and the blue petal flowers did NOT breed true. Based on that, calculate the values of  $p$ ,  $p^2$ ,  $2pq$ ,  $q$  and  $q^2$ .

For this problem, you need to know what the recessive phenotype is. Because the black petal flowers “true breed” (meaning when they are self-crossed, they produce offspring ONLY like themselves and have black petal flowers), they HAVE to be homozygous recessive or “aa”.

$$q^2 = 1853 \text{ black petals} / 5281 \text{ total flowers} = 0.351$$

$$q = 0.592$$

$$p = 0.408$$

$$p^2 = 0.166$$

$$2pq = 0.483$$

$$\text{CHECK: } (0.166 + 0.483 + 0.351) = 1.0$$

Determine the allele frequency of the current group of individuals, calculate the expected genotypic frequencies of their offspring and the frequency of each phenotype.

10. A flock of 342 scarlet crows, of which 102 individuals in the flock have a green colored legs (a dominant mutation).

$$342 - 102 = 240 \text{ recessive phenotypes}$$

$$240/342 = 0.702 = q^2$$

$$q = 0.838$$

$$p = 0.162$$

$$\text{CHECK: } (0.162 + 0.838 = 1.0)$$

$$p^2 = 0.026$$

$$2pq = 0.272$$

$$\text{CHECK: } (0.026 + 0.272 + 0.702 = 1.0)$$

"A" phenotype ( $A_{-}$ , green legs) =  $p^2 + 2pq = 0.026 + 0.272 = 0.298$  (342 birds) = 102 green legged birds

"a" phenotype (aa, normal legs) = 0.702 (342 birds) = 240 normal legged birds

11. A flock of 1,200 seagulls with 867 homozygous dominant individuals.

$$867/1200 = 0.723 = p^2$$

$$p = 0.850$$

$$q = 0.150$$

$$q^2 = 0.022$$

$$2pq = 0.255$$

$$\text{CHECK: } (0.723 + 0.255 + 0.022 = 1.0)$$

"A" phenotype =  $p^2 + 2pq = 0.723 + 0.255 = 0.978$  (1200 cowbirds) = 1174 dominant trait

"a" phenotype = 0.022 (1200 cowbirds) = 26 recessive trait birds

12. A colony of adult lab mice is variable for the length of their tail. They have been carefully bred so that the mice are all homozygous, some dominant, some recessive. Long tails are dominant to short tails. 25 have long tails and 75 have short tails.

A. Calculate p and q for this population.

25 long tails, so 50 "L" alleles

75 short tails, so 150 "l" alleles

$$\text{LONG allele frequency} = p = 50/(50 + 150) = 0.25 = p$$

$$\text{SHORT allele frequency} = q = 150/(50 + 150) = 0.75 = q$$

$$\text{CHECK: } (0.25 + 0.75 = 1.0)$$

B. Someone decided that they should be "liberated" from their cages and let all of the mice out. They bred randomly. If 400 mouse pups are born as a result, predict how many of the offspring had long tails and how many had short tails.

$$p^2 = 0.0625$$

$$2pq = 0.375$$

$$q^2 = 0.5625$$

$$\text{CHECK: } (0.0625 + 0.375 + 0.5625 = 1.0)$$

$$\text{"A" phenotype} = 0.0625 + 0.375 = 0.4375 \times 400 = 175 \text{ long-tailed pups (dominant trait)}$$

$$\text{"a" phenotype} = 0.5625 \times 400 = 225 \text{ short-tailed pups (recessive trait)}$$

13. What are the allele frequencies in an isolated field of 382 pink, 355 white and 103 red snapdragons. Flower colors in snapdragons exhibit incomplete dominance.

RED = RR

PINK = Rr

WHITE = rr

where R = red allele and r = white allele

Remember, each flower has 2 alleles. Calculate the number of alleles for each type.

RED alleles =  $(103 \times 2) + 382 = 588$  red alleles

\*the pink flowers contribute 382 red alleles

WHITE alleles =  $(355 \times 2) + 382 = 1,092$  white alleles

\*the pink flowers contribute 382 red alleles

TOTAL flowers = 840, so there is a TOTAL of 1680 alleles

$$\text{RED} = 588/1680 = 0.35 \text{ red allele frequency}$$

$$\text{WHITE} = 1,092/1680 = 0.65 \text{ white alleles frequency}$$

# THE HARDEST PROBLEMS

Read carefully and be thoughtful.

14. I walked into a room that was full of 750 palmetto bugs. Out of these 750 palmetto bugs, 35 of them were albinos and SUPER scary! These albinos were so scary and mean looking that I squashed and killed them with my shoe and let the remaining palmetto bugs breed. The next generation of baby palmetto bugs totaled 15123. Assuming that the allele for albinism is recessive, **how many of these babies would be ALBINO?**

Before selection,  $35/750 = 0.047 = q^2$

$q = 0.217$

$p = 0.783$

$p^2 = 0.613$

$2pq = 0.340$

CHECK:  $(0.613 + 0.340 + 0.047 = 1.0)$

$AA = 750 \times 0.613 = 459.8 = 460$  normal palmetto bugs

$Aa = 750 \times 0.340 = 255$  normal palmetto bugs

CHECK:  $460 + 255 = 715$  normal palmetto bugs and 35 albino palmetto bugs;  $715 + 35 = 750$  total bugs

KILL 35 albinos, which leaves 715 palmetto bugs (either AA or Aa, or 1430 alleles) left over

After selection, 255/1430 alleles are "a",  $q = 0.178$

$q^2 = 0.0317$

$0.0317 \times 15123$  baby palmetto bugs = 479.4 = **479 albino baby palmetto bugs** (Error in rounding, accept  $\pm 5$  or **474 to 484** albino babies)

15. Texas Wesleyan University used to have a colony of gerbils that were used for genetics research. I was particularly interested in two coat color genes. One gene causes gerbils to be brown (B-) or black (bb). The other is called dilution and it causes colors to look lighter. For example, genetically black gerbils that are also dd look grey, genetically brown but dd gerbils are called "cinnamon". In 1997, I had the following gerbils:

First, you need to understand the GENOTYPES of the different phenotypes. B=brown allele, b=black allele, B>b, D is the allele to express full color, D>d, dd dilutes the color

|             |      |
|-------------|------|
| 35 grey     | bbdd |
| 24 black    | bbD_ |
| 22 cinnamon | B_dd |
| 12 brown    | B_D_ |

A high school teacher gave me two litters of her gerbils. They were

|            |      |
|------------|------|
| 5 brown    | B_D_ |
| 3 cinnamon | B_dd |
| 5 black    | bbD_ |
| 3 grey     | bbdd |

- A. What was the frequency of the B and b alleles of my colony after the new gerbils were added?

$(35 \text{ grey} + 24 \text{ black}) + (3 \text{ grey} + 5 \text{ black}) = 67 \text{ bb mice}$

$$67/(109 \text{ total mice}) = 67/109 = 0.6147 = q^2$$

$$q \text{ (or recessive "b" allele)} = \mathbf{0.784} \text{ and } p \text{ (or dominant "B" allele)} = \mathbf{0.216}$$

B. What was the frequency of the D and d alleles of my colony after the new gerbils were added?

$$(35 \text{ grey} + 22 \text{ cinnamon}) + (3 \text{ cinnamon} + 3 \text{ grey}) = 63 \text{ dd mice}$$

$$63/(109 \text{ total mice}) = 63/109 = 0.578 = q^2$$

$$q \text{ (or recessive "d" allele)} = \mathbf{0.760} \text{ and } p \text{ (or dominant "D" allele)} = \mathbf{0.240}$$

16. The gene for color-blindness is on the X chromosome. Men have one X chromosome (and a Y). Women have two X chromosomes. I have personally tested a large number of students for color-blindness and have found that, on average, one male will be color-blind in a class of 30. Assuming that such a class is 1/2 male and 1/2 female, how many women would I have to test before I found one who was color-blind?

$$15 \text{ males, } 1/15 = 0.0667 = q$$

$$q^2 = 0.0044$$

$$44/10,000 \text{ women are color-blind or } 1 \text{ in } 227.3 \text{ is color blind.}$$

On average, you would need to test approximately **227 women** before you found one who was color blind.

17. Many cattle breeders obtain bulls of one breed to sire calves with cows of another breed. One of the most common crosses is black angus bulls with hereford cows. Purebred herefords all have a white face that is caused by a dominant allele. As a result of being inbred you can assume that they are all homozygous. Purebred black angus never have a white face. What percentage of the offspring (from crossing black angus bulls with hereford cows) in such a herd will have a white face?

$$AA \times aa = \mathbf{100\% Aa}$$

## TOO MUCH TIME ON YOUR HANDS?

19. If you really think you understand how these problems work, enjoy a challenge and have nothing else to do for a while try Question X!

The City of Angels has a population of 106. Their blood types are:

$$B = 4.5 \times 10^5$$

$$O = 3.6 \times 10^5$$

$$A = 1.3 \times 10^5$$

$$AB = 6.0 \times 10^4$$

What are the allele frequencies in this population?